

Mid Semester Examination 2025

Course: B.Tech. (Mechanical Engineering and Aerospace Engineering)

Semester: 3rd

Paper Name: Engineering Mathematics-III

Paper Code: TMA - 303

Time: 1:30 Hours

MM: 50

Note:

- (1) Answer all the questions by choosing any one of the sub questions.
- (2) Each question carries 10 marks.

Q. 1

(10 Marks)

- (a) Construct the partial differential equation from $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$, where α is a parameter.

CO-1

OR

- (b) Using Lagrange's method to solve the given partial differential equation $xzp + yzq = xy$.

CO-1

Q. 2

(10 Marks)

- (a) Solve the partial differential equation $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \sin x$.

CO-1

OR

- (b) Solve $(D_x - D_y - 1)(D_x - D_y + 2)z = e^{2x-y} + 2x$, where $D_x \equiv \frac{\partial}{\partial x}$, $D_y \equiv \frac{\partial}{\partial y}$.

CO-1

Q. 3

(10 Marks)

- (a) Using Newton's method, find a real root of an equation $xe^x = 1$ correct to three decimals.

CO-2

OR

- (b) Solve $4 \frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 16 \log(x + 2y)$.

CO-1

Q. 4

(10 Marks)

- (a) Solve $\frac{dy}{dx} = -y$ numerically from $x = 0$ to $x = 1$ using explicit forward first order Euler method with initial condition of $y(0) = 1$ and step size (h) of 0.2. Find the absolute value of error in $y(1)$ calculated using analytical and numerical solution in percentage (%) (calculate the error using analytical solution as the basis and use three decimal places).

CO-2

OR

- (b) Using Runge-Kutta method to obtain y for $x = 0.1$ correct to four decimal places. Given that the initial value problem (IVP) $\frac{dy}{dx} = x + y$ with initial condition $y(0) = 1$.

CO-2

Q. 5

(10 Marks)

- (a) A river is 80 metres wide. The depth (y) of the river at a distance (x) from bank to given in the following table:

x	0	10	20	30	40	50	60	70	80
y	0	4	7	9	12	15	14	8	3

Using Trapezoidal rule find the approximate area of cross-section of the river.

CO-2

OR

- (b) A car is moving at the speed of 30 m/sec. Suddenly brakes are applied. The speed of the car per second after ' t ' seconds is given below:

Time (t)	0	5	10	15	20	25	30	35	40	45
Speed (v)	30	24	19	16	13	11	10	8	7	5

Using Simpson's $\frac{3}{8}$ rule to compute the distance moved by the car in 45 seconds.

CO-2